

Survival Analysis of Heart Dataset

STAT 432/532

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Outline

- Introduction
- Relevant Background
- Data and Model Description
- Model Comparison
- Limitations
- Conclusions

Dataset Introduction

- The heart data is found in the survival package
- Consists of 4 different datasets:
 - Jasa
 - Jasa1
 - Heart
 - Stanford2

Source

J Crowley and M Hu (1977), Covariance analysis of heart transplant survival data. *Journal of the American Statistical Association*, **72**, 27–36.

Covariance Analysis of Heart Transplant Survival Data

JOHN CROWLEY and MARIE HU*

This paper presents a number of analyses to assess the effects of various covariates on the survival of patients in the Stanford Heart Transplantation Program. The data have been updated from previously published versions and include some additional covariates, such as measures of tissue typing. The methods used allow for simultaneous investigation of several covariates and provide estimates of the relative risk of transplantation as well as significance tests.

KEY WORDS: Heart transplant; Survival data; Covariance analysis.

When more than one potential recipient is matched on blood type for a given donor heart, other criteria, such as the availability of the patient and recently, tissue typing, have been used. In a few cases a patient was passed over until a pulmonary infection, which might cause complications after transplantation, had cleared. However, in general it is thought that no serious systematic bias has been introduced by this selection of patients for the transplant operation. In fact, the belief

Biological Introduction

- Heart transplants are performed when other treatments for heart problems have not shown improvement in health.
- In adults, heart failure can be caused by many reasons, including:
 - Coronary artery disease
 - Cardiomyopathy (weakening of the heart muscle)
 - Congenital heart defect (being born with a heart problem)
 - Many others
- In children, heart failure is often caused by cardiomyopathy and congenital heart defects.
- Many factors such as age and previous medical history are considered when being put on the waitlist to receive a heart transplant.

Other Relevant Introduction

- Computational tools:
 - Analysis is done using solely R
 - Some key libraries: survival, ggplot2, survminer
- Exploratory Data Analysis is via Data Visualization
- Goal for today is to build a simple model and assess its limitations and conclude with adjustments with a final Cox-Proportional Hazard model with time-dependent covariates

Data

Data

▶ heart	172 obs. of 8 variables
▶ jasa	103 obs. of 14 variables
▶ jasa1	170 obs. of 8 variables
▶ stanford2	184 obs. of 5 variables

	birth.dt	accept.dt	tx.date	fu.date	fustat	surgery	age	futime	wait.time	transplant	mismatch	hla.a2	mscore	reject
1	1937-01-10	1967-11-15	NA	1968-01-03	1	0	30.844627	49	NA	0	NA	NA	NA	NA
2	1916-03-02	1968-01-02	NA	1968-01-07	1	0	51.835729	5	NA	0	NA	NA	NA	NA
3	1913-09-19	1968-01-06	1968-01-06	1968-01-21	1	0	54.297057	15	0	1	2	0	1.11	0
4	1927-12-23	1968-03-28	1968-05-02	1968-05-05	1	0	40.262834	38	35	1	3	0	1.66	0
5	1947-07-28	1968-05-10	NA	1968-05-27	1	0	20.785763	17	NA	0	NA	NA	NA	NA
6	1913-11-08	1968-06-13	NA	1968-06-15	1	0	54.595483	2	NA	0	NA	NA	NA	NA
7	1917-08-29	1968-07-12	1968-08-31	1970-05-17	1	0	50.869268	674	50	1	4	0	1.32	1
8	1923-03-27	1968-08-01	NA	1968-09-09	1	0	45.349760	39	NA	0	NA	NA	NA	NA
9	1921-06-11	1968-08-09	NA	1968-11-01	1	0	47.162218	84	NA	0	NA	NA	NA	NA
10	1926-02-09	1968-08-11	1968-08-22	1968-10-07	1	0	42.502396	57	11	1	2	0	0.61	1

JASA DATA - PACKAGE: SURVIVAL

Variables from Raw Data: (Jasa)

birth.dt: birth date

accept.dt: acceptance into program

tx.date: transplant date

fu.date: end of followup

fustat: dead or alive

surgery: prior bypass surgery (0 = no, 1 = yes)

age: age (in years)

futime: followup time

wait.time: time before transplant

transplant: transplant indicator (0 = no, 1 = yes)

mismatch: mismatch score

hla.a2: particular type of mismatch

mscore: another mismatch score

reject: rejection occurred

CONTAINS THE RAW DATA OF SURVIVAL OF PATIENTS ON THE WAITING LIST FOR THE STANFORD HEART TRANSPLANT PROGRAM

Data

Data

▶ heart	172 obs. of 8 variables
▶ jasa	103 obs. of 14 variables
▶ jasa1	170 obs. of 8 variables
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	start	stop	event	age	year	surgery	transplant	id
1	0.0	50.0	1	-17.15537303	0.12320329	0	0	1
2	0.0	6.0	1	3.83572895	0.25462012	0	0	2
3	0.0	1.0	0	6.29705681	0.26557153	0	0	3
4	1.0	16.0	1	6.29705681	0.26557153	0	1	3
5	0.0	36.0	0	-7.73716632	0.49007529	0	0	4
6	36.0	39.0	1	-7.73716632	0.49007529	0	1	4
7	0.0	18.0	1	-27.21423682	0.60780287	0	0	5
8	0.0	3.0	1	6.59548255	0.70088980	0	0	6
9	0.0	51.0	0	2.86926762	0.78028747	0	0	7
10	51.0	675.0	1	2.86926762	0.78028747	0	1	7

HEART DATA - PACKAGE: SURVIVAL

Variables from Processed Data: (Heart)

start, stop, event: Entry and exit time and status for this interval of time

age: age - 48 years

year: year of acceptance (in years after 1 Nov 1967)

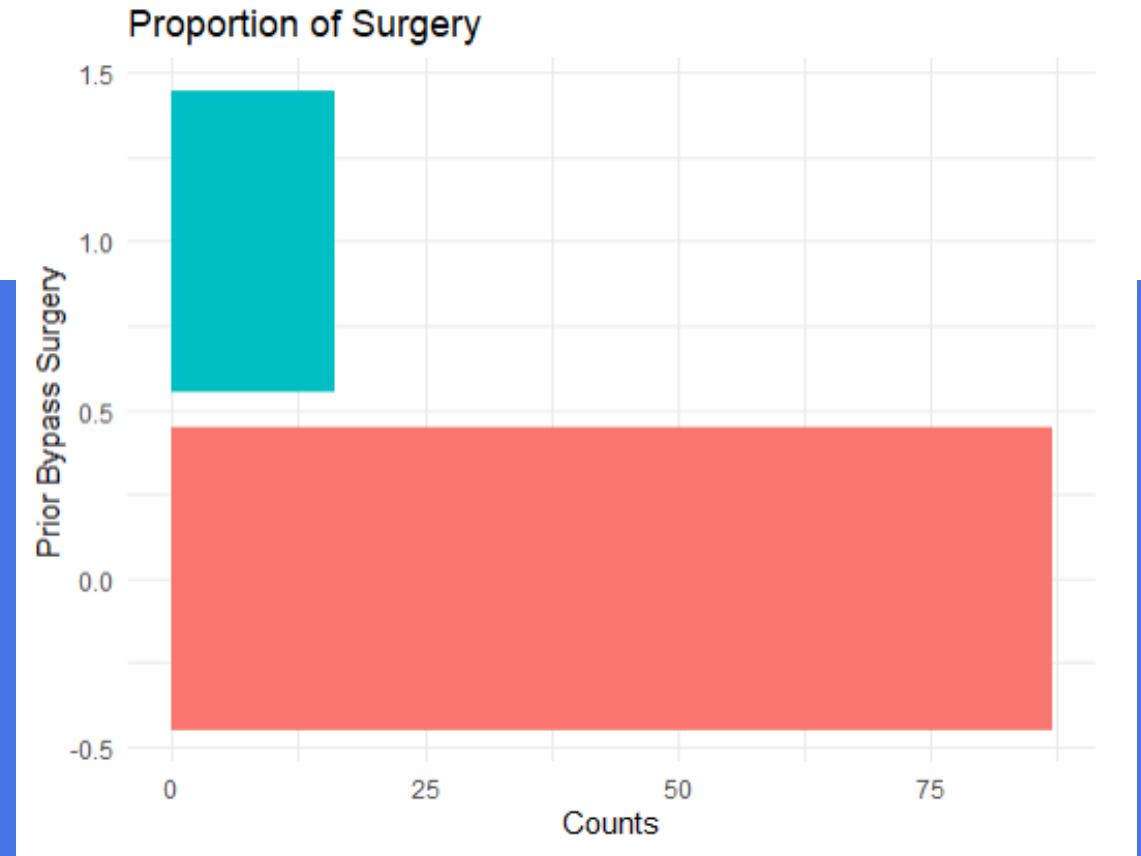
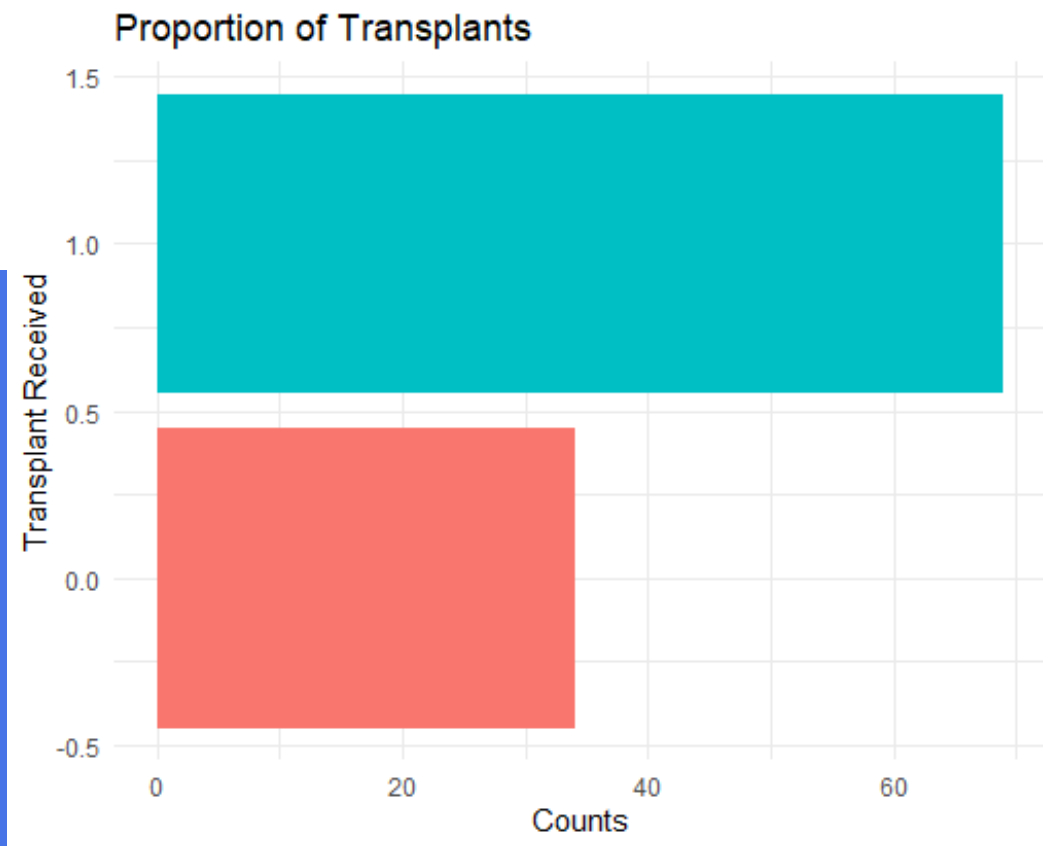
surgery: prior bypass surgery 1=yes

transplant: received transplant 1=yes

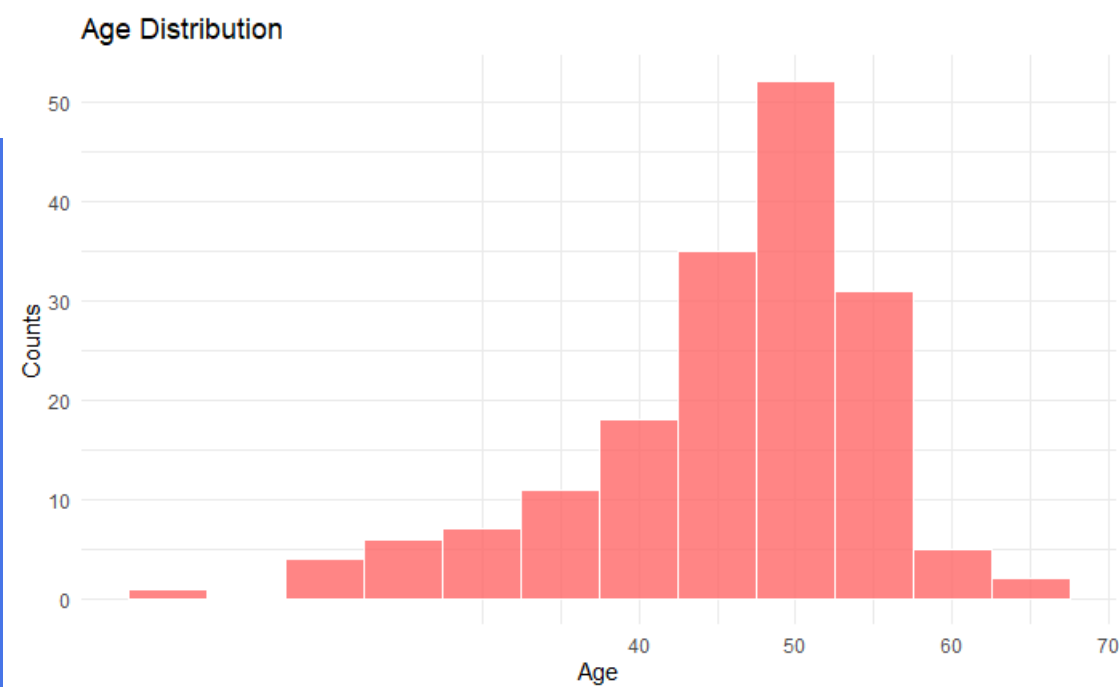
id: patient id

CONTAINS THE PROCESSED DATA OF SURVIVAL OF PATIENTS ON THE WAITING LIST FOR THE STANFORD HEART TRANSPLANT PROGRAM

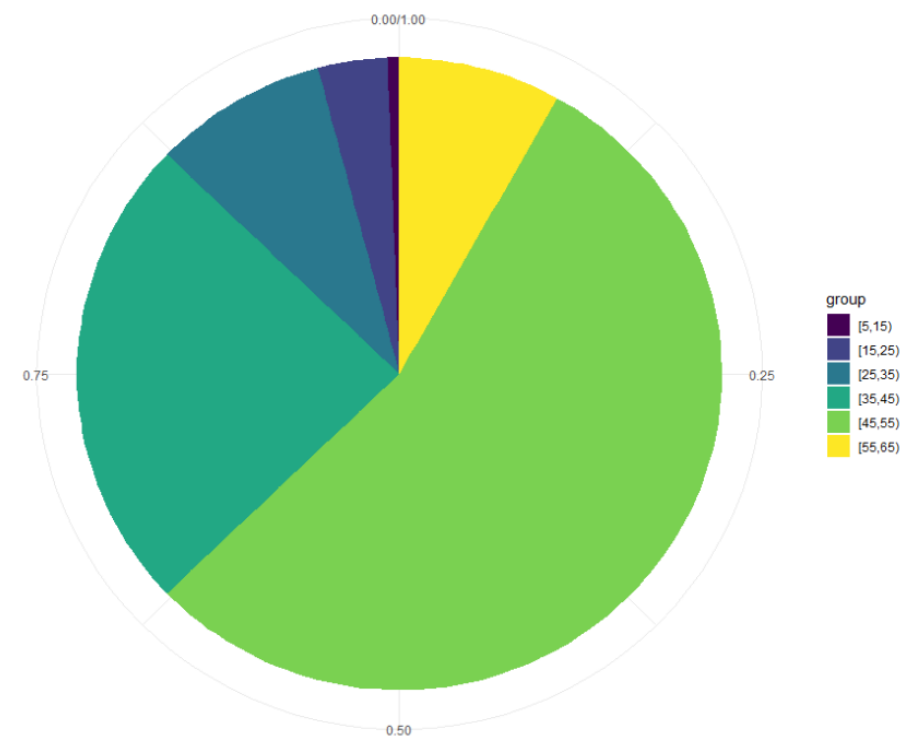
Data



Data



Proportion of Age Groups



Data

- Very small dataset, only $n = 103$ individuals
- Some variables are VERY unbalanced!
 - 34 did not receive transplant, 69 did receive transplant
 - 87 did not receive prior bypass surgery, 16 did receive prior bypass surgery
 - age
- Lots of missing values in original jasa dataset
 - Will be using the processed data (heart)
- Censored data due to death, drop out, end of study, may lead to unidentifiability issues (i.e. We do not know the cause of potential bias)

Basic Introduction

- Survival function: Probability of an individual surviving beyond time x

$$S(X) = P(X > x)$$

- Estimator of the survival function: Product-Limit Estimator (Kaplan-Meier Estimator)

$$\hat{S}(t) = \begin{cases} 1 & \text{if } t < t_1 \\ \prod_{t_i \leq t} \left[1 - \frac{d_i}{y_i} \right] & \text{if } t_1 \leq t \end{cases}$$

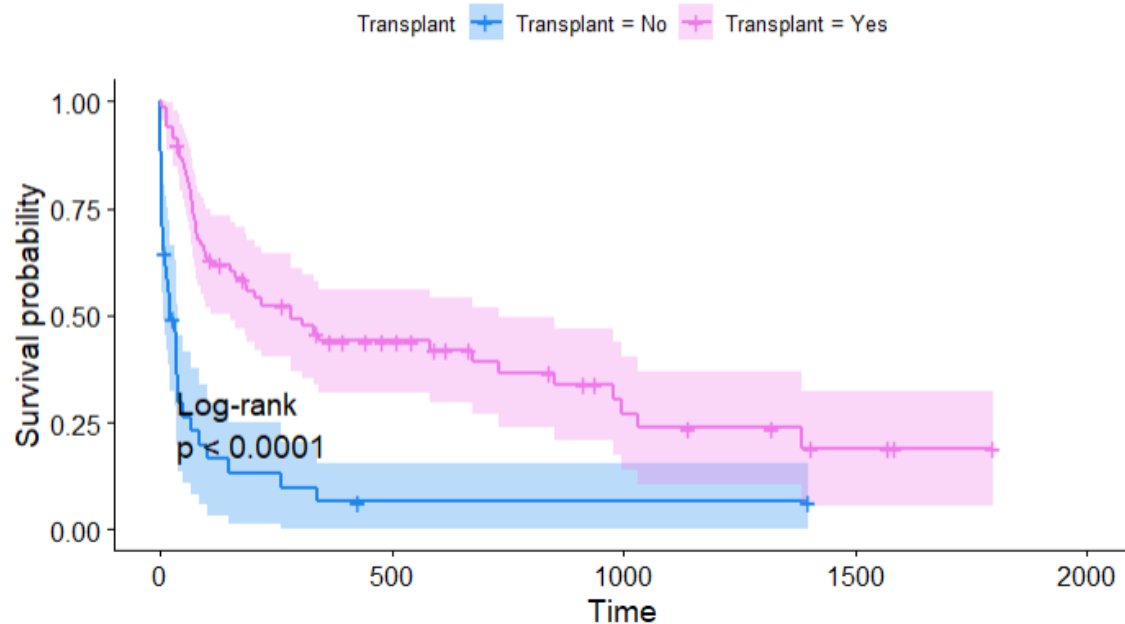
where d_i/Y_i is the estimate of the conditional probability that an individual who survives just prior to time t_i experiences the event at time t_i

Basic Model, Log-Rank Test

- Want to examine survival as a function of whether subject received a heart transplant, or has a history of prior bypass surgery
- Hypothesis:
 - H_0 : There are no differences in the K-M Estimates between groups
 - H_A : There are differences in the K-M Estimates between groups
- Compute p-value of log-rank test directly by indicating $pval = T$ with *ggsurvplot* in R

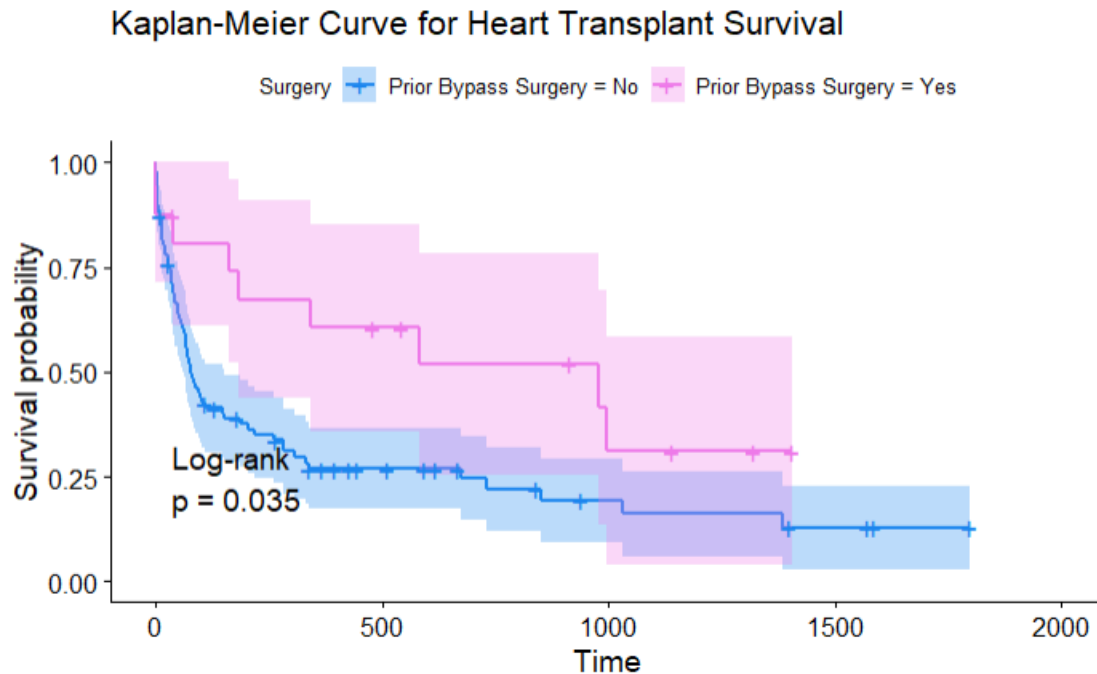
Survival Probability between Transplant Groups

Kaplan-Meier Curve for Heart Transplant Survival



- Computed the survival probabilities between individuals who have received a transplant and individuals who have not received a transplant.
- From the plot, we can see that those who have received a heart transplant indeed have a higher survival probability
- Log-rank test p-value < 0.0001.

Survival Probability between Surgery Groups



- Computed the survival probabilities between individuals who have/have not underwent prior bypass surgery
- From the plot, we can see that those who have went under bypass surgery previously indeed have a higher survival probability
- Log-rank test p-value < 0.035 .



Limitations

Briefly, patients are admitted to the Stanford program (given personal and family consent) after a medical conference at which it is decided that the patient is not likely to respond to other forms of therapy. A donor heart, matched on blood type, is then sought; this search has lasted anywhere from a few days to almost a year. Some patients die before a suitable heart is found. The total number of patients waiting when a heart is found is generally one or two, though it has been as high as seven.

- Does not capture the change of a variable whose value may be different over time
- “The key rule for time dependent covariates in a Cox model is simple and essentially the same as that for gambling: you cannot look into the future. A covariate may change in any way based on past data or outcomes, but it may not reach forward in time.”
 - <https://stat.ethz.ch/R-manual/R-devel/library/survival/doc/timedep.Rnw>

Cox Proportional Hazard model

- One of many popular regression techniques for survival analysis
- Hazard rates are proportional – looking at two individuals with covariates Z and Z^* , the ratio of their hazards is free of the time t .
- Important Assumption! Ratio of hazards is constant over time
- $h(t|Z) = h_0(t)\exp \sum_{k=1}^p \beta_k Z_k]$
 - $h(t)$ is the hazard at time t
 - $h_0(t)$ is the baseline hazard rate
 - p number of Z predictors
- Referring to the limitations from the previous slide, we should consider including time dependent covariates in addition to the fixed covariates (defined using indicator variable)

Cox Proportional Hazard Model

- Use processed heart data (Still bad! Why?):

call:

```
coxph(formula = surv(start, stop, event) ~ age + year + surgery +  
      transplant, data = heart)
```

	coef	exp(coef)	se(coef)	z	p
age	0.02717	1.02754	0.01371	1.981	0.0476
year	-0.14635	0.86386	0.07047	-2.077	0.0378
surgery1	-0.63721	0.52877	0.36723	-1.735	0.0827
transplant1	-0.01025	0.98980	0.31375	-0.033	0.9739

Likelihood ratio test=15.11 on 4 df, p=0.004476
n= 172, number of events= 75

	chisq	df	p
age	0.43657	1	0.51
year	1.05790	1	0.30
surgery	0.00994	1	0.92
transplant	0.12665	1	0.72
GLOBAL	2.11630	4	0.71

Cox Proportional Hazard Model

- Use processed heart data (Still bad! Why?):

```
call:
coxph(formula = Surv(start, stop, event) ~ age + year + surgery +
      transplant, data = heart)

n= 172, number of events= 75
```

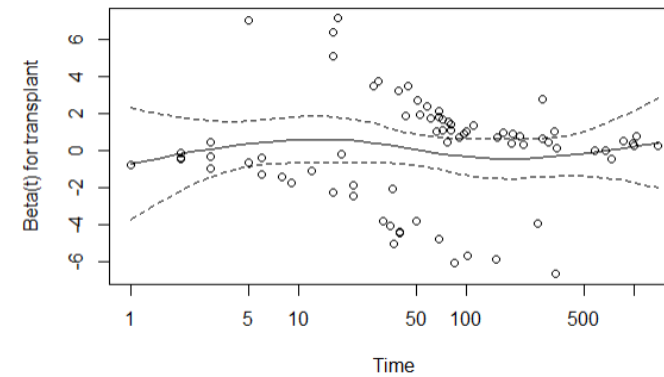
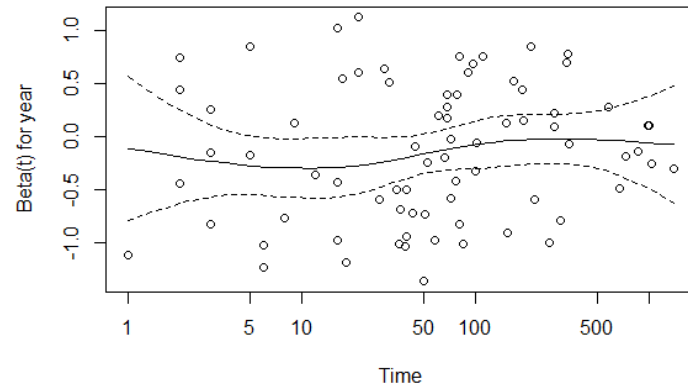
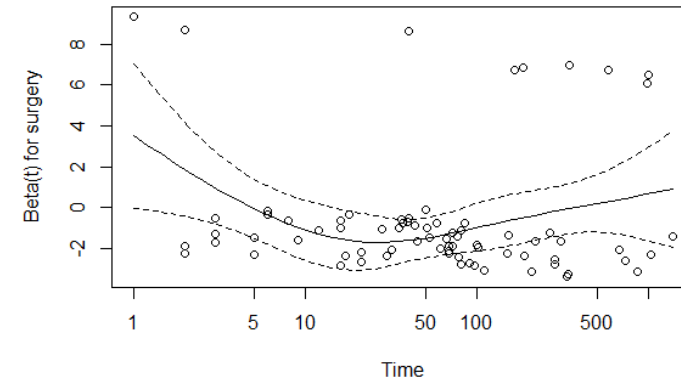
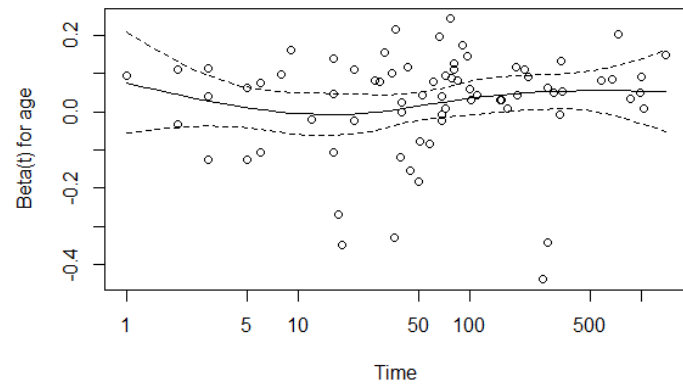
	coef	exp(coef)	se(coef)	z	Pr(> z)
age	0.02717	1.02754	0.01371	1.981	0.0476 *
year	-0.14635	0.86386	0.07047	-2.077	0.0378 *
surgery1	-0.63721	0.52877	0.36723	-1.735	0.0827 .
transplant1	-0.01025	0.98980	0.31375	-0.033	0.9739

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0275	0.9732	1.0003	1.0555
year	0.8639	1.1576	0.7524	0.9918
surgery1	0.5288	1.8912	0.2574	1.0860
transplant1	0.9898	1.0103	0.5352	1.8307

Concordance= 0.636 (se = 0.033)
Likelihood ratio test= 15.11 on 4 df, p=0.004
Wald test = 14.49 on 4 df, p=0.006
Score (logrank) test = 15.03 on 4 df, p=0.005

Cox Proportional Hazard Model



Cox Proportional Hazard Model

- A better model to consider would be to include an interaction of year and age with transplant. As well, stratify the transplant variable as it is time dependent.
- Biological reasonings may be that:
 - If two individuals of the same condition, same factors (except one – in this case age) had a chance at the donor heart, I speculated that maybe the heart will go to the younger individual
 - maybe a higher chance of survival if you are younger
 - If an individual is placed on the donor list before another, the first person typically gets the transplant first
 - “moving up the transplant list”

Cox Proportional Hazard Model

- Transplant is time dependent! Age and year have interactions

call:

```
coxph(formula = surv(start, stop, event) ~ strata(transplant) +  
      age + year + surgery + age:transplant + year:transplant,  
      data = heart)
```

	coef	exp(coef)	se(coef)	z	p
age	0.01981	1.02001	0.01792	1.105	0.2690
year	-0.26754	0.76526	0.10692	-2.502	0.0123
surgery1	-0.64205	0.52621	0.37234	-1.724	0.0846
age:transplant1	0.03004	1.03050	0.02805	1.071	0.2842
year:transplant1	0.23723	1.26774	0.14310	1.658	0.0974

Likelihood ratio test=19.39 on 5 df, p=0.001626
n= 172, number of events= 75

	chisq	df	p
age	0.1153	1	0.73
year	0.0441	1	0.83
surgery	0.1084	1	0.74
age:transplant	1.7515	1	0.19
year:transplant	1.1689	1	0.28
GLOBAL	5.4202	5	0.37

Cox Proportional Hazard Model

```
call:
coxph(formula = surv(start, stop, event) ~ strata(transplant) +
      age + year + surgery + age:transplant + year:transplant,
      data = heart)
```

```
n= 172, number of events= 75
```

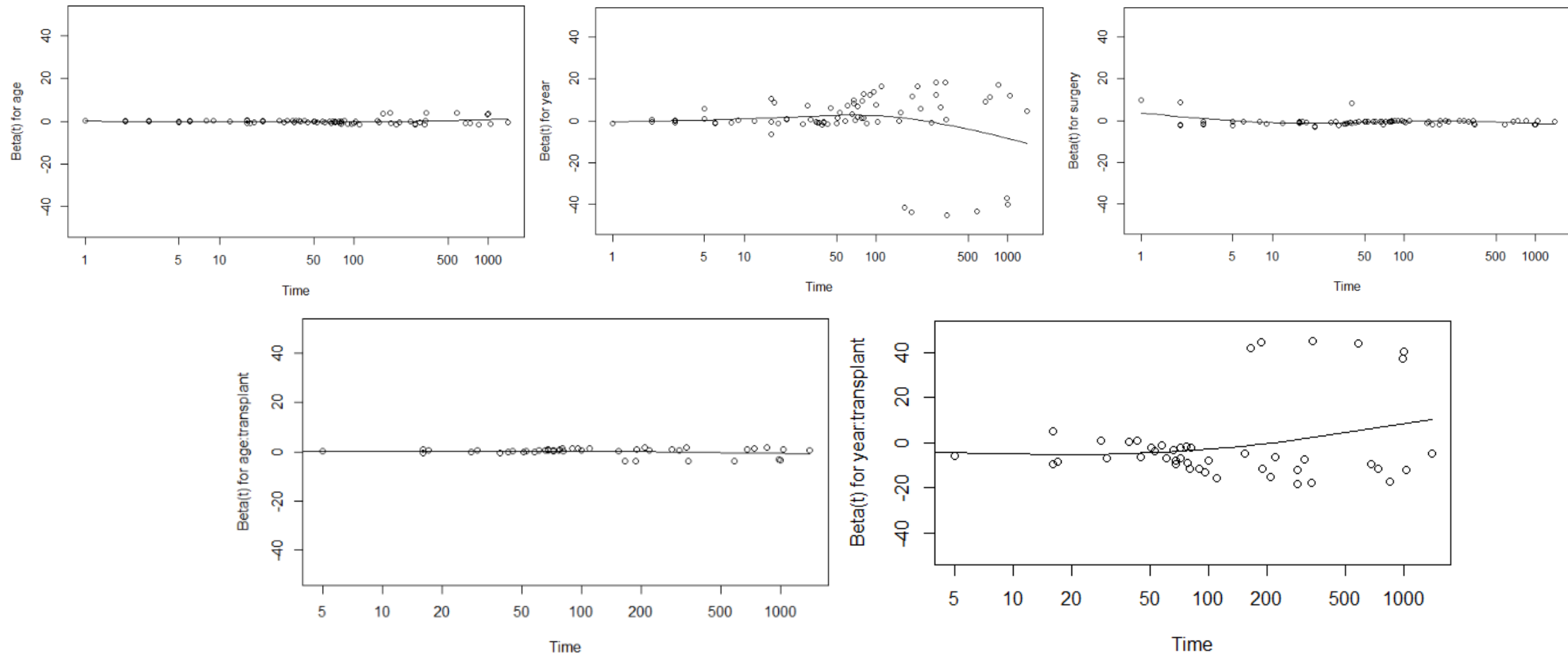
	coef	exp(coef)	se(coef)	z	Pr(> z)
age	0.01981	1.02001	0.01792	1.105	0.2690
year	-0.26754	0.76526	0.10692	-2.502	0.0123 *
surgery1	-0.64205	0.52621	0.37234	-1.724	0.0846 .
age:transplant1	0.03004	1.03050	0.02805	1.071	0.2842
year:transplant1	0.23723	1.26774	0.14310	1.658	0.0974 .

```
---
signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

	exp(coef)	exp(-coef)	lower .95	upper .95
age	1.0200	0.9804	0.9848	1.0565
year	0.7653	1.3067	0.6206	0.9437
surgery1	0.5262	1.9004	0.2536	1.0917
age:transplant1	1.0305	0.9704	0.9754	1.0887
year:transplant1	1.2677	0.7888	0.9577	1.6782

```
Concordance= 0.645 (se = 0.036 )
Likelihood ratio test= 19.39 on 5 df, p=0.002
Wald test               = 17.95 on 5 df, p=0.003
Score (logrank) test = 18.79 on 5 df, p=0.002
```

Cox Proportional Hazard Model



Summary

From simple K-M Estimate we were able to see

- Those who received a transplant had a higher chance of survival compared to those who have not
- Those who underwent bypass surgery had a higher chance of survival compared to those who have not

From Proportional Cox-Hazard Model we were able to see:

- It is important to consider how the variables are defined in the study
- Biological interpretations are important to consider when accounting for interactions and figuring out what the time dependent covariates are



THANK YOU

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References

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